

ENGINEERING HANDOFF SUPPLEMENT

ESP-FC Firmware Delta Handoff

Post-baseline firmware changes only

Repository: Deepak7769/ESP-FC

Current master: cf4c237baa24eb401d969aa13c39a656e03ffc48

Date: 27 September 2026

Scope

This document intentionally excludes the hardware, airframe, dual-ESP32, LoRa/iBUS, powertrain, perfboard, old GPS and general project background already documented in the previous technical handoff. It records only the firmware work completed after that baseline.

Executive delta

- Angle V2 is now the only authoritative Angle-mode outer controller and is always compiled.
- AltHold V2 and LAND V2 are implemented in the shared Assisted V2 architecture, with guarded compile-time activation policy.
- Estimator freshness, receiver validation, arming, failsafe, mixer/output and sensor failure handling were hardened.
- Dedicated native and ESP32 validation environments were added; current GitHub CI is green across every configured target.

Version anchors

Current repo HEAD cf4c237b...e03ffc48

GitHub Actions run 36306884990 - SUCCESS

Runtime config reference final CLI.txt - build 7bf8293 (25 Sep 2026)

Important: the CLI snapshot is older than current master and is not proof that cf4c237b is the binary presently flashed on hardware.

ARCHITECTURE

Shared Assisted V2 control policy

Angle is authoritative; AltHold/LAND remain guarded.

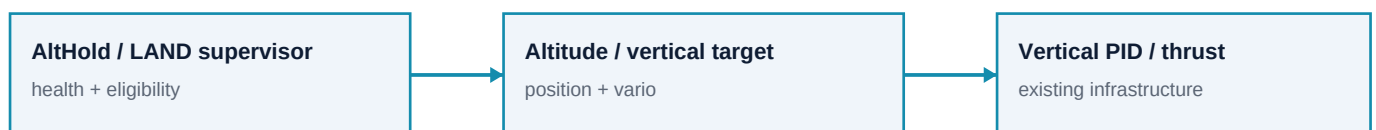
Feature-selection contract

Path	Policy	Meaning
Angle V2	Always compiled	Only Angle outer controller; no legacy Angle fallback.
AltHold V2	Feature-gated	Enabled through Assisted V2 policy/test macros.
LAND V2	Feature-gated	Requires AltHold V2 and reuses Angle V2 leveling.
Production policy	Explicit acknowledgement	ESPFC_ASSISTED_V2_ACTIVE outside safe bench requires ESPFC_ASSISTED_V2_OUTPUT_ACK.
Centered stick	Configurable channel	Candidate integration uses input index 6 (AUX3).

High-level control topology



Guarded vertical path



State model split

- AngleV2State now owns Roll/Pitch angleTarget, rateTarget, pScale and active state.
- AssistedModeShadowState now carries AltHold/LAND vertical state only; Angle state was separated from the old shadow structure.
- Controller transition state is explicit: Angle entry, AltHold entry and Assisted update timing are tracked independently.

CONTROLLER DELTA

Angle V2 is now authoritative

One attitude outer loop feeding the existing rate PID.

Control law and transition behavior

- On Angle entry, the target is initialized from the measured Roll/Pitch attitude. This prevents an instantaneous jump to level.
- Requested angle is derived from the existing level angle-limit configuration and pilot Roll/Pitch input.
- angleTarget moves toward requested angle with a 120 deg/s slew limit.
- Outer-loop gain is LEVEL P multiplied by angleV2.pScale; the generated rateTarget is clamped to the configured level rate limit.
- After the first nominal cycle, Assisted dt is measured and clamped to 0.00025..0.050 s to prevent a stalled/debugged cycle creating a large one-shot target jump.
- When Angle V2 is inactive, targets track measured attitude and its Roll/Pitch rateTarget is cleared.

Authoritative path

```
pilot -> requested angle -> slew-limited angleTarget -> angle error -> LEVEL P x pScale  
-> rateTarget -> rate clamp -> setpoint.rate -> existing inner rate PID -> mixer
```

Health and failure semantics

- Attitude freshness threshold is 100000 us. Actuator health also requires active gyro + accel, healthy attitude state, finite Euler/quaternion values and a plausible quaternion norm.
- If the Angle switch remains ON and the estimator fails, an Angle fault latch removes the mode. Re-entry requires a deliberate OFF->ON switch cycle.
- If Angle is requested but Angle V2 cannot provide a valid active target, Roll/Pitch setpoint rates are zeroed. The removed legacy Angle outer controller is not used as fallback.

Scaler and configuration migration

- Legacy outerPid scaling was removed. ACT_ANGLE_P now occupies bit 4. Bits 5..7 remain reserved so persisted scaler numeric layout does not shift.
- CLI names remain pid_level_p, pid_level_angle_limit and pid_level_rate_limit. No separate angle_v2_* CLI namespace was added.

Reference config only: level P=200, angle limit=30 deg, rate limit=200 deg/s; Roll 35/45/15/45, Pitch 35/45/15/50, Yaw 35/45/0/40.

VERTICAL CONTROL

AltHold V2 rework

Guarded controller path with independent estimator freshness.

Entry, state and ownership

- AltitudeState now has its own healthy flag and lastUpdateUs, independent of attitude/barometer freshness.
- AltHold becomes active only with healthy altitude estimation, fresh attitude and a fresh valid barometer sample.
- On entry it captures current estimated altitude and current vertical speed for bumpless transfer.
- The configured centered-stick channel must exist, be valid and be centered for entry. Once active, stick deflection is a normal climb/descent command rather than a fault.
- Estimator/channel failure latches AltHold fault state and requires OFF->ON before re-entry.

Current implementation constants

These are code constants in current HEAD, not universal tuning recommendations.

Quantity	Current value	Role
Altitude position Kp	0.50	Converts altitude error to vertical-rate correction.
Target error window	+/- 2.0 m	Prevents unreachable target backlog / position windup.
Position correction	+/- 1.0 m/s	Bounds position-loop contribution.
Descent cap	-1.0 m/s	Vertical request lower bound.
Climb cap	+1.5 m/s	Vertical request upper bound.
Vertical acceleration	1.0 m/s ²	Slew-limits vertical-rate target.
LAND descent request	-0.50 m/s	Used when LAND owns vertical command.

Repository policy status

AltHold V2 exists and is exercised by dedicated safe-bench/native environments, but it is still controlled by the Assisted V2 compile-time activation policy rather than being an unconditional production path.

FAILSAFE SUPERVISOR

LAND V2 and termination hardening

Shared leveling and vertical-control infrastructure.

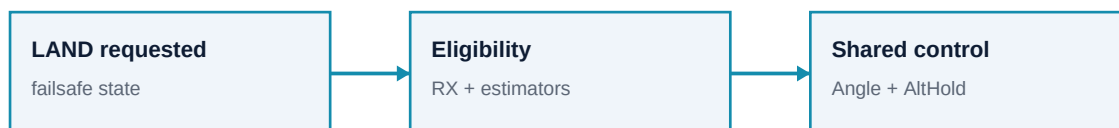
LAND architecture

- LAND reuses the same Angle V2 leveling controller; it does not create a second competing attitude loop.
- LAND shares the AltHold V2 altitude/vertical-rate infrastructure for descent control.
- The logical LAND output starts blocked. Eligibility requires an active LAND request, a previously valid receiver state and healthy assisted estimators.
- If estimator eligibility is lost during active LAND, assisted output is blocked/terminated rather than continuing from stale state.
- LAND fault diagnostics remain latched across request termination, so the failure reason is not erased immediately.

Touchdown and recovery changes

- Touchdown qualification was hardened with ground-relative altitude, low-thrust qualification and confirmation hysteresis/timing.
- LAND-entry thrust is retained as a touchdown reference.
- Recovery/termination was made terminal and bumpless; a new arm clears the previous LAND latch.
- Switch-triggered failsafe pilot-input handling and receiver-recovery qualification were hardened.

Supervisor flow



Eligibility lost -> block / terminate assisted LAND; preserve diagnostic latch.

CROSS-CUTTING DELTA

Receiver, arming, output and sensor hardening

Area	Change	Why it matters
Receiver / input	Reject unavailable AUX channels; complete/stable PPM frames; distinguish SBUS dropped frames from link loss; harden timing/frame-rate recovery.	Prevents default/stale channel contents from becoming control intent.
Arming / failsafe	Stale gyro blocks arm and disarms; invalid/non-finite attitude fails small-angle check; GPS arming flag clears when feature disabled; startup/recovery failsafe logic qualified.	Arming and recovery now depend on current valid state.
Mixer / output	Disarm output propagation decoupled from normal control cycle; invalid mixer entries/counts fail closed; persisted output pulse ranges sanitized; stale RPM telemetry expires.	Reduces stale command / stale telemetry retention.
Estimator / sensors	Independent altitude freshness; rejected fusion does not reintegrate stale world acceleration; baro correction independent of IMU rate; reload preserves history; first pressure sample primed; rejected height does not corrupt vario.	Fixes the major estimator defects reported in the earlier comparison.
Sensor fail-closed	BMP280 initialization errors and HMC5883L configuration/overflow paths fail closed.	Device failures no longer masquerade as valid data.
Protocol / other	CRSF MSP response truncation/underflow fixed; telemetry bounds and buzzer-event handling hardened.	Removes parser/state edge cases.

Failure-handling pattern

Unavailable, stale, invalid or out-of-range state is rejected or expired; control-mode entry is qualified; output paths fail closed; recovery requires validated conditions rather than zero-initialized/default values.

GAP CLOSURE

Earlier review findings versus current HEAD

Based on Firmware comparison.txt versus cf4c237b.

Earlier gap	Current state
Angle V2 was shadow-only	Angle V2 is the authoritative Roll/Pitch outer loop.
AltHold V2 had no thrust authority	Guarded active AltHold V2 path is implemented and tested.
No independent altitude freshness	AltitudeState has independent health/timestamp; freshness regression passes.
Rejected fusion could reuse stale acceleration	Rejected-fusion regression now verifies no reintegration.
Estimator reload reset history	Reload-preserves-history regression now passes.
Pressure filter started from zero	First-sample priming regression now passes.
Baro correction depended on IMU rate	Variable-dt and correction-independence regressions now pass.
Changed disarm could wait behind control cycle	Disarm-output propagation regression now passes.

Representative regression coverage now present

- fusion rejects invalid accel without poisoning state; rejected fusion does not reintegrate world acceleration.
- barometer vario used only once per sample; rejected height sample does not correct vario.
- complementary variable-dt behavior and altitude correction independence from IMU rate.
- altitude reload history preservation, pressure-filter priming, independent altitude freshness and disarm-output propagation.

VALIDATION

Dedicated environments and current results

Environment	Key flags / role	Purpose
native_angle_v2	Native C++17, UNIT_TEST, SAFE_BENCH	Angle V2 regression suite.
esp32_angle_v2_bench	ESP32 + SAFE_BENCH	Target compile; physical ESC driver not initialized/attached.
native_assisted_v2	Native + SAFE_BENCH + AltHold/LAND test macros	Guarded assisted controller regression.
esp32_assisted_v2_bench	ESP32 + SAFE_BENCH + AltHold/LAND test macros	Target-side guarded assisted build.
esp32_assisted_v2_candidate	ESP32 SAFE_BENCH + generic Assisted V2 + channel 6	Compiles production-policy candidate without physical motor attach.
native_assisted_v2_active	Native UNIT_TEST + Assisted active + OUTPUT_ACK + channel 6	Validates production activation policy without hardware.

Observed validation results

LOCAL: 79 / 79 PASS

ESP32 BENCH BUILD: SUCCESS

GitHub Actions run 36306884990 on cf4c237b: SUCCESS

Jobs passed: test, esp32, esp32s2, esp32s3, esp32c3, esp8266, rp2040 and rp2350.

Migration note

A temporary preprocessor error around landingV2OwnsControl was introduced during warning cleanup and immediately corrected. It is not present in current HEAD.

CONFIGURATION SNAPSHOT

Older CLI/runtime reference - not current HEAD identity

Source snapshot reports build 7bf8293, 25 Sep 2026.

Do not use this snapshot as proof that cf4c237b is flashed.

It is included only to give a new developer the last supplied runtime/config context.
Version/status should be checked before comparing hardware logs to the current repository.

Observed runtime snapshot

Item	Snapshot value
Gyro clock	8000 Hz
Gyro rate	4000 Hz
Control loop	4000 Hz
Mixer rate	4000 Hz
Accel	500 Hz
Barometer	50 Hz
Magnetometer	100 Hz
Detected devices	MPU6500/SPI, BMP280/I2C, QMC5883P/I2C
Fusion	Mahony; P=50, I=5; magnetometer fusion disabled
Output config	PWM async, 400 Hz

Important delta from old handoff

The older handoff assumed an approximately 125 Hz processing loop. This later runtime snapshot reports a 4 kHz control/mixer loop. Treat this as an observed build/config result, not a guarantee for every target or future configuration.

Snapshot mode map: MODE_ARMED on input index 4 high range; MODE_AIRMODE and MODE_ANGLE on input index 5 high range.

HANDOFF MAP

Where a new developer should look first

File / area	What changed / why it matters
Control/AssistedModeV2.h	Feature selection and activation policy.
ModelState.h	AngleV2State, estimator freshness and assisted/failsafe state.
ModelConfig.h	Flight modes, scaler bit layout, ACT_ANGLE_P.
Control/Controller.h/.cpp	Angle authority; AltHold/LAND control; transition state.
Control/Actuator.cpp	Mode qualification, fault latches, scaler routing, LAND supervisor, arming safety.
Output/Mixer.cpp	SAFE_BENCH physical driver interlock and output handling.
Control/Altitude.hpp + AHRS	Estimator timing/history corrections.
SensorManager.cpp + BaroSensor.cpp	Fusion rejection and pressure handling.
Input parsers	PPM/SBUS/CRSF and channel-validation hardening.
platformio.ini + CI workflow	Dedicated validation environments and target matrix.
test/test_fc/test_fc.cpp	Angle, AltHold, LAND, estimator, receiver, failsafe and mixer regressions.

Current status

GREEN: ANGLE V2

GREEN: CI

GUARDED: ALTHOLD/LAND

- Established: current repo compiles/tests in CI; Angle V2 is integrated and authoritative; supplied basic Angle V2 regression suite passes.
- Guarded: AltHold V2 and LAND V2 exist, but activation remains compile-time policy controlled.
- Not established by supplied sources: real-world behavior of Sep 27 HEAD; whether the currently flashed hardware binary exactly equals cf4c237b.
- Not yet present as dedicated tests: explicit time-bounded slew, negative-stick symmetry, separate two-direction clamp cases, controller-level non-finite attitude case, and end-to-end Angle scaler gain-effect tests.

ORIENTATION

Selected post-baseline change timeline

When	Firmware delta
26 Sep	AltHold/LAND safe-bench authority; touchdown state; shared activation policy; production-policy regressions; centered-stick entry gate.
26 Sep	Receiver/channel validation; failsafe termination/recovery; stale gyro/disarm; mixer fail-closed; pulse-range/RPM telemetry hardening.
27 Sep early	LAND interlocks/diagnostics; low-thrust/touchdown refinements; sensor fail-closed changes; CRSF MSP fix.
27 Sep migration	AngleV2State; ACT_ANGLE_P scaler migration; Actuator/Controller moved off legacy outerPid; Angle V2 made authoritative; regression updates.
Current	cf4c237b; GitHub Actions run 36306884990 fully green.

Software-only handoff checklist

- Verify the checkout is cf4c237b or later before using this document as the code reference.
- Run native_angle_v2 regression tests and the esp32_angle_v2_bench compile after modifying Angle/assisted code.
- Inspect AssistedModeV2.h before changing AltHold/LAND activation policy.
- Do not reintroduce legacy outerPid references; preserve ACT_ANGLE_P bit 4 and reserved bit positions 5..7.
- Keep Angle V2 as the single Angle-mode outer-loop authority unless the architecture is deliberately redesigned and re-tested.
- Confirm hardware version/status before comparing flight-controller logs to this Sep 27 repository state.
- Treat regression failures as defects to investigate; do not silently change controller constants merely to make tests pass.

Handoff summary

The major architectural milestone after the previous handoff is the move from shadow/legacy assisted logic to a single authoritative Angle V2 outer loop, plus guarded AltHold/LAND V2 and broad fail-closed hardening. Current source and CI are internally consistent at cf4c237b.